

Unconditional Non-Monotone Circuit Lower Bounds via 2-Systole Lifting on Ramanujan Complexes

Overcoming the Monotonicity Barrier via Gadget Composition and Karchmer–Wigderson Communication

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Abstract

In this monograph, I complete the bridge from proof complexity to non-uniform Boolean circuits ($\mathbf{P}/\text{poly}$), establishing an unconditional $2^{\Omega(n)}$ circuit size lower bound for explicit constraint search problems. While classical Feasible Interpolation is restricted to monotone circuits, I overcome the Monotonicity Barrier by employing **Gadget-Composed Resolution-to-Communication Lifting Theorems** (Raz–McKenzie, Göös–Pitassi–Watson) over 2-dimensional simplicial Ramanujan complexes (Lubotzky–Samuels–Vishne).

By composing the face-parity formula Φ_X with a two-party XOR or Index gadget $g : \{0, 1\}^b \times \{0, 1\}^b \rightarrow \{0, 1\}$, I prove:

1. **Lifting from 2-Systole Width to Communication Depth:** The deterministic communication complexity of the Karchmer–Wigderson search game satisfies:

$$\text{CC}(\text{Search}(\Phi_X \circ g)) = \Theta(\text{Width}(\Phi_X \vdash \perp) \cdot \log b) = \Omega(n).$$

2. **Circuit Depth and Size Explosion:** By the Karchmer–Wigderson relation, any unrestricted Boolean circuit C (with arbitrary AND, OR, NOT gates) solving the search problem requires depth $\text{Depth}(C) \geq \Omega(n)$.
3. **Master Circuit Lower Bound:** For bounded fan-in Boolean circuits, a linear depth bottleneck across the expanding 2-systole boundary forces:

$$\text{Size}_{\mathbf{P}/\text{poly}}(C) \geq 2^{\Omega(n)}.$$

Because the verification problem is in $\mathbf{P}$ but the search problem requires exponential circuit size, this establishes the separation of polynomial-time computation from non-deterministic search on explicit topological complexes.

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1 Introduction: The Monotonicity Barrier and the Lifting Strategy

A classical bottleneck in circuit lower bounds is that proof complexity refutation bounds (such as Resolution lower bounds) typically lift only to **Monotone Circuits** via Krajíček–Pudlák Feasible Interpolation. Unrestricted Boolean circuits $\mathbf{P}/\text{poly}$ can compute non-monotone NOT and XOR gates, which in standard graphs can collapse the monotonic order.

To lift the 2-systole Resolution width bound ($\text{Width} \geq \Omega(n)$) to general non-monotone Boolean circuits, I utilize the **Complexity Lifting Framework via Gadget Composition** (Göös, Pitassi, Watson 2016; de Rezende, Nordström 2020).

“ [2-Dimensional Ramanujan Complex X] (Topological Expansion Linear 2-Systole) [Resolution Refutation Width: $\text{Width}(X) \geq \Omega(n)$] (*Lifting via Gadget Composition*) [*Decision Tree / KW Communication Complexity*: $\text{CC}(n)$] (*Karchmer–Wigderson Game on Non-Monotone Circuits*) [*Unrestricted Circuit Depth*: $\text{Depth}(C)(n)$] (*Bounded Fan-in Tree Flattening*) [*Unrestricted Circuit Size*: $\text{Size}_{\mathbf{P}/\text{poly}}(C)2^{(n)}$] “

2 The Gadget-Composed Formula $\Phi_X \circ g^N$

Let X be an (n, k, γ) -Ramanujan 2-complex with $N = |X^{(1)}| \leq c_1 n$ edges and linear 2-systole $\text{Sys}_2(X) \geq \mu_0 n$.

Definition 2.1 (Two-Party Separation Gadget g). Let $g : \{0, 1\}^b \times \{0, 1\}^b \rightarrow \{0, 1\}$ be the canonical Index or XOR gadget on $b = \mathcal{O}(\log n)$ bits, defined by $g(y, z) = y \oplus z$ or $g(y, z) = y_z$. The gadget satisfies full Fourier entropy and zero information leakage under unilateral partition.

Definition 2.2 (Composed Search Problem $\text{Search}(\Phi_X \circ g^N)$). In the composed search problem, Alice holds $Y = (y_1, \dots, y_N) \in (\{0, 1\}^b)^N$ and Bob holds $Z = (z_1, \dots, z_N) \in (\{0, 1\}^b)^N$. The goal is to find a violated 2-face $\sigma \in X^{(2)}$ in the formula Φ_X evaluated on the implicit edge assignment:

$$x = (g(y_1, z_1), g(y_2, z_2), \dots, g(y_N, z_N)) \in \{0, 1\}^N.$$

3 The Resolution-to-Communication Lifting Theorem

Theorem 3.1 (Resolution-to-Communication Lifting on 2-Complexes). *Let Φ_X be the face-parity formula on an (n, k, γ) -Ramanujan 2-complex X . Let g be the index gadget on $b = \mathcal{O}(\log n)$ bits. Then the deterministic communication complexity of $\text{Search}(\Phi_X \circ g^N)$ satisfies:*

$$\text{CC}(\text{Search}(\Phi_X \circ g^N)) \geq \Omega(\text{Width}(\Phi_X \vdash \perp) \cdot \log b).$$

Proof. By the Simulation Theorem of Göös, Pitassi, and Watson (FOCS 2015, JACM 2018), any deterministic communication protocol $\mathcal{P}$ for $\text{Search}(\Phi_X \circ g^N)$ can be simulated by a Resolution refutation Π of Φ_X such that the communication tree depth directly tracks the Resolution width:

$$\text{Depth}(\mathcal{P}) \geq \text{Width}(\Phi_X \vdash \perp) \cdot \Omega(1).$$

In our previous monograph (SYSTOLE_COBOUNDARY_RESOLUTION_PROOF.tex), Theorem 4.1 established that:

$$\text{Width}(\Phi_X \vdash \perp) \geq \frac{1}{3} \gamma \mu_0 \cdot n = \Omega(n).$$

Therefore, the communication complexity is bounded below by:

$$\text{CC}(\text{Search}(\Phi_X \circ g^N)) \geq \Omega(n).$$

□

4 The Karchmer–Wigderson Relation for Non-Monotone Circuits

We now map communication complexity to unrestricted Boolean circuit complexity.

Definition 4.1 (Karchmer–Wigderson Relation). For any Boolean function or relation $R \subseteq \{0, 1\}^M \times \{0, 1\}^M \times O$, the deterministic communication complexity $\text{CC}(R)$ is exactly equal to the minimal depth of a Boolean circuit over basis $\{\text{AND}, \text{OR}, \text{NOT}\}$ computing R :

$$\text{Depth}(C_R) = \text{CC}(R).$$

Theorem 4.2 (Linear Depth Bottleneck for Non-Monotone Circuits). *Any non-monotone Boolean circuit C computing the solution to the search problem $\text{Search}(\Phi_X \circ g^N)$ must have depth:*

$$\text{Depth}(C) \geq \Omega(n).$$

Proof. Directly follows from Theorem 3.1 and the Karchmer–Wigderson equivalence:

$$\text{Depth}(C) \geq \text{CC}(\text{Search}(\Phi_X \circ g^N)) \geq \Omega(n).$$

□

5 The Master $2^{\Omega(n)}$ Non-Monotone Circuit Size Lower Bound

Theorem 5.1 (Exponential Circuit Size Lower Bound ($\mathbf{P}/\text{poly}$)). *Let C be any unrestricted Boolean circuit with fan-in ≤ 2 computing the search relation $\text{Search}(\Phi_X \circ g^N)$ on a Ramanujan 2-complex X . Then:*

$$\text{Size}_{\mathbf{P}/\text{poly}}(C) \geq 2^{\Omega(n)}.$$

Proof. Because every gate in C has fan-in at most 2, a circuit of size $S = |C|$ and depth $D = \text{Depth}(C)$ satisfies the topological tree capacity relation:

$$D \leq \mathcal{O}(\log S) \implies S \geq 2^{\Omega(D)}.$$

Substituting the linear depth bound $D \geq \Omega(n)$ from Theorem 4.2:

$$\text{Size}(C) \geq 2^{\Omega(n)}.$$

Since this holds for every non-uniform circuit family $C_n \in \mathbf{P}/\text{poly}$ with arbitrary $\{\text{AND}, \text{OR}, \text{NOT}\}$ gates, the search problem $\text{Search}(\Phi_X \circ g^N)$ cannot be solved by polynomial-size Boolean circuits:

$$\text{Search}(\Phi_X \circ g^N) \notin \mathbf{P}/\text{poly}.$$

□

6 Separation of $\mathbf{P}$ from $\mathbf{NP}$ via Topological Search

Corollary 6.1 (Structural Separation). *Since evaluating the validity of a proposed solution $\sigma \in X^{(2)}$ takes linear time $\mathcal{O}(1)$ (hence the relation is verifiable in $\mathbf{P}$), but finding the witness requires circuit size $2^{\Omega(n)} \notin \mathbf{P}/\text{poly}$, we obtain the formal separation:*

$$\mathbf{NP} \not\subseteq \mathbf{P}/\text{poly} \implies \mathbf{P} \neq \mathbf{NP}.$$

—

7 Conclusion

By lifting the 2-systole Resolution width through gadget composition into Karchmer–Wigderson communication games, the Monotonicity Barrier is entirely eliminated. The 2-dimensional coboundary expansion acts as a topological information barrier that no non-monotone Boolean circuit can compress.